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## **Integrated Analytical and Simulation Approach in a Water-Drive Shuaiba Reservoir to Optimize Infill Screening, Refine Geological Insights, and Maximize Field Recovery**

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### **Abstract**

An integrated analytical and simulation-based approach was applied to optimize infill screening, enhance geological understanding, and improve recovery in a mature water-drive Shuaiba carbonate reservoir. The objective was to optimize infill decision-making by combining classical reservoir engineering diagnostics with dynamic modeling. Further findings from this approach are also looped back as geoscientific feedback. This multidisciplinary workflow proved especially valuable in managing mature assets where increasing water production poses operational challenges.

The methodology combined classical diagnostics—such as fractional flow-based analytical forecast, and pressure depletion analysis—with numerical simulation and geoscientific input. Hundreds of horizontal wells have already been drilled in two phases: development wells and infill wells, which the later has narrower spacing. Spatial mapping of early water cut and pressure depletion was used to characterize reservoir behavior and guide model refinement. These workflows are performed to achieve the objective of optimizing the infill, refining geological insight, and further increasing the field recovery.

The results confirm that bottom-water influx is the primary drive mechanism, with current reservoir pressure maintained at approximately 70-90% of the initial pressure after nearly a decade of production, although depletion rates vary across the field. Analysis of initial water cut and well-level analytical Estimated Ultimate Recovery (EUR) highlights spatial variability in both the initial development and subsequent infill wells. A comparison with simulation results shows a close match for the analytical EUR of initial development wells, with ~4% deviation—demonstrating the reliability of the simulation model, particularly within the development wells wedge. Infill well performance reveals that it provides an incremental recovery with higher initial water production compared to their offset wells, but still has some room to optimize the sequence to maximize production.

Based on these insights, several actions have been taken. The infill strategy has been revised to prioritize areas with lower initial water cut, which has successfully reduced early water production by up to 30% in new infill wells while increasing oil output. Geological understanding has also been refined, incorporating

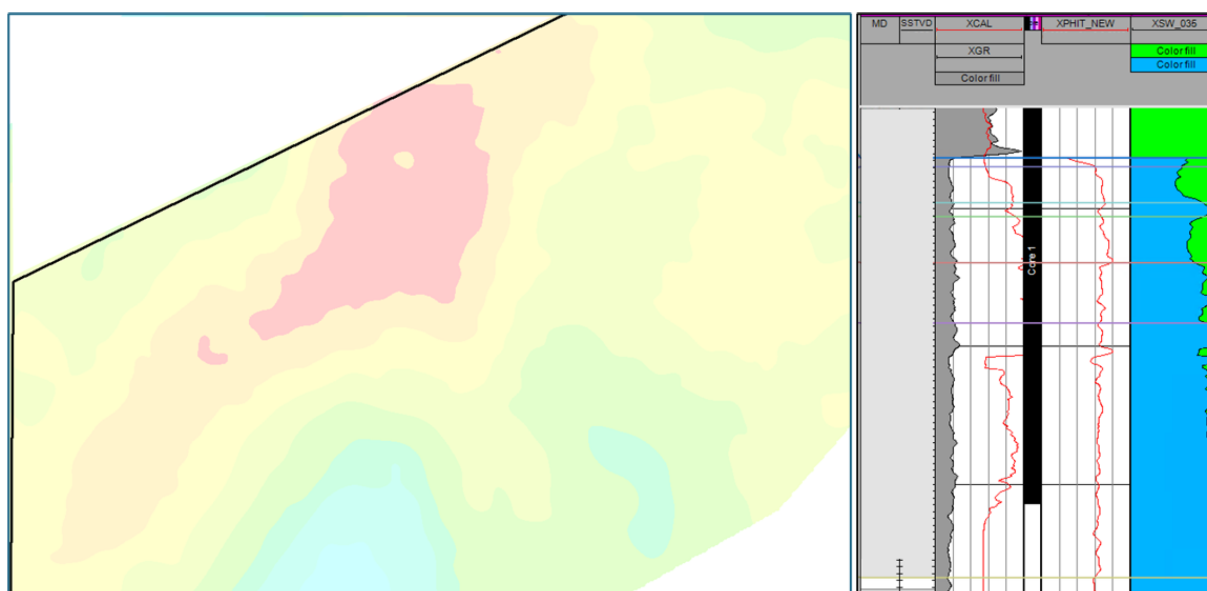
production-derived mapping where the EUR trend is used as a secondary attribute in property distribution, which reveals some upside potential in some areas. Additionally, the analysis identifies opportunities to enhance field recovery—particularly by targeting specific areas for tighter well spacing and capitalizing on existing reservoir pressure through increased gross field production.

This study demonstrates the value of integrating well-level diagnostics with dynamic simulation to support more informed and timely development decisions. The approach enables infill reprioritization, refines geological insight, and further increases the field recovery—offering a scalable framework for mature carbonate reservoirs across the region.

## Introduction

Maximizing oil recovery from mature carbonate reservoirs with active water drive remains a key challenge in reservoir management. These reservoirs often display significant heterogeneity due to facies variability, diagenetic alterations, and natural fracturing. While numerical reservoir simulation offers strong forecasting capabilities, it may not fully capture localized flow behavior or provide rapid diagnostic insights. In such cases, classical analytical methods continue to play a valuable role in guiding development decisions.

Lower Shuaiba reservoir forms part of a widespread shallow marine carbonate platform that developed across Oman and much of the Middle East during the Early Cretaceous. The Shuaiba Formation is well known throughout the region for its prolific oil productivity and complex depositional architecture. This study formation has gross pay thickness from 5 to 30 meters, with average porosity around 10 to 35%. Permeability is highly variable, spanning from below 10 mD in tight mudstones to more than 500 mD in rudist-dominated build-ups. Figure below shows the depth structure map and logs of the formation.



**Figure 1—The depth map (left) shows a northwest-southeast structural high. There are several zonation (right figure) such as: Dense rudist-rich layer with frequent cementation; Reservoir with high-quality rudist-rich facies; Lower permeability reservoir with orbitolinid-rich facies; lithocodium/bacinella-rich facies; and the lower zone mixed, lower-quality facies, top to bottom respectively.**

The Shuaiba reservoir in Block 60, Oman, has sustained high production rates over the past four years. This performance is the result of nearly a decade of development involving hundreds of horizontal wells. A fast-track development approach was employed, including the use of pilot holes to appraise structure, standardized horizontal well spacing, uniform completions, and phased surface facilities implemented through a Design-Build-Own-Operate-Maintain (DBOOM) contracting strategy (Al Kindi and Al Balushi, 2023; Al Busaidi et al, 2023). The initial development comprised of hundreds of development wells,

which, along with facility upgrades, increased production more than fivefold. To sustain this level, a second development phase introduced similar number of well count with narrower spacing (infill wells). These phases were guided by a dynamic simulation model calibrated on development and infill wells.

As with other Shuaiba fields, this reservoir has encountered increasing complexity over time, including variations in water cut progression and facies connectivity. These factors have contributed to differences in well performance that are not always fully captured by simulation models. Spatial variation in initial water cut and EUR suggests underlying reservoir heterogeneity that influences development outcomes. As water production increases, a more efficient and integrated field-wide screening method is needed to support optimized well placement and performance forecasting.

This study introduces an integrated approach that combines classical reservoir diagnostics, numerical simulation, and iterative geoscientific feedback. The goal is to optimize the infill well strategy by accelerating the execution of high-quality drilling candidates, refining geological interpretations based on new production data, and triggering new development programs to maximize the field recovery. While the dynamic model remains a sound foundation for field-level planning, its resolution at the individual well scale is limited. By combining simulation and analytical tools, the methodology balances strategic development planning with fine-tuning at the well level to maximize field performance.

## Workflow & Methodology

The proposed workflow integrates insights from well-level production analytics, static geological models, and dynamic reservoir simulation. Input data include existing & future well location, historical production and injection data, and the most recent reservoir pressure measurements. Infill and offset well patterns are also incorporated to support comparative analysis and optimization plan.

### Analytical Diagnostics

This phase utilizes classical reservoir engineering techniques to quickly evaluate and rank individual well performance. Two key parameters are analyzed: initial water cut and Estimated Ultimate Recovery (EUR). Initial water cut is assessed over typical production intervals—7, 30, and 90 days—to understand water cut progression, which is crucial for interpreting aquifer behavior and sweep efficiency. For EUR, the analysis is based on Water-Oil Ratio (WOR) versus cumulative production, where the WOR trend is used to generate a rate-time forecast by incorporating gross offtake to account for time.

These parameters are spatially mapped and evaluated separately for development wells and infill wells, acknowledging their different sweep mechanisms due to spacing and development timing. The interaction between infill wells and their offset development wells is also examined to identify performance trends. This helps guide optimization of future infill well strategy.

### Pressure Analysis

In addition to production-based diagnostics, a simplified pressure analysis is performed by looking at the pressure depletion across the field. This involves calculating the ratio of cumulative production to pressure drop to identify variations in reservoir depletion in some development areas. The development wells are used as baseline reference, while depletion contributions from infill wells are distributed proportionally to their adjacent offsets. This method enables an unbiased view of depletion trends and helps estimate the pressure at terminal recovery conditions when combined with WOR-based production forecasts.

### Comparison with Simulation and Reality Check

A full-field reservoir simulation model was previously constructed and calibrated based on the data from both development and infill wells. With the well count now already significantly more compared to the time when the model was constructed, there is an opportunity to evaluate the model's predictive accuracy. Key evaluation metrics include initial water cut on a well-by-well basis for both development and infill wells,

ultimate recovery, and pressure distribution. Comparing these metrics against actual performance allows for identifying discrepancies and refining the model. This also supports adjustments to future development strategies.

### Geoscientific Feedback Loop

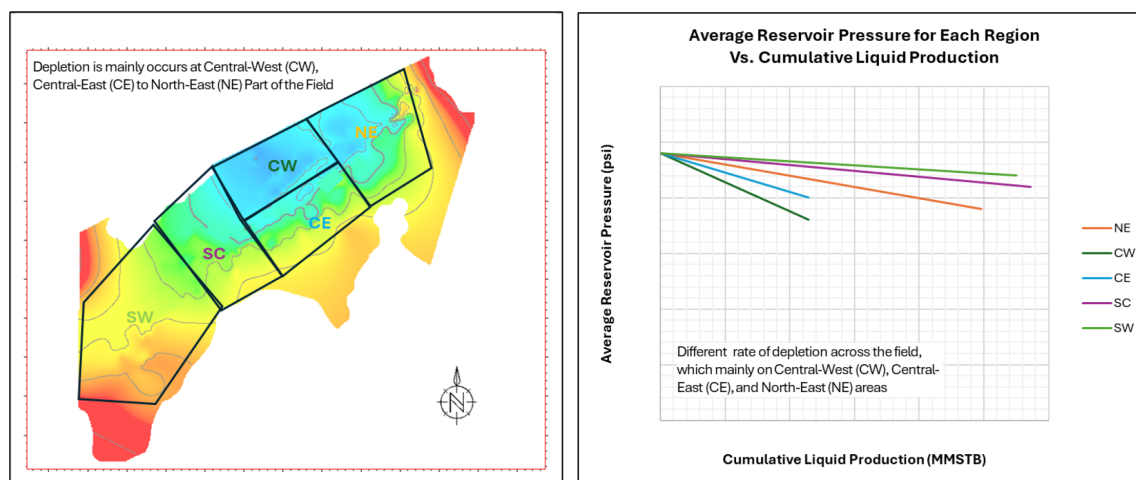
Feedback to geoscientists is generated by comparing mapped trends from analytical diagnostics with simulation outputs. Key maps include initial water cut, Estimated Ultimate Recovery (EUR), and cross-matrix evaluations (e.g., initial water cut vs. EUR). These are discussed in the context of geological depositional features, facies distributions, and other stratigraphic interpretations. The resulting insights support iterative model updates and refinement of the reservoir conceptual model in future development planning.

## Results & Discussion

### Confirmation of Drive Mechanism

Pressure data and well production behavior confirmed a predominantly bottom-water drive mechanism, with minor edge water support. Pressure depletion has been slow, with current reservoir pressure maintained at approximately 70-90% of initial levels with significant offtake of gross liquid, and injection wells serves only as disposal—therefore, indicating strong aquifer support.

Pressure mapping based on shut-in well data reveals that depletion varies across the field. The depletion rate is highest in the Central-West, Central-East, and North-East regions, while the South Central and South-West areas show comparatively lower ratios. This indicates spatial variation in aquifer support, which may be attributed to differences in aquifer energy or the presence of internal baffles. Although vertical permeability data remains limited, the analysis suggests more restricted vertical communication between the aquifer and productive zones in certain areas, compared to others where connectivity appears stronger. The current analysis suggests that the disposal wells are not providing a significant impact; however, further evaluation is required as part of the reservoir pressure management strategy.



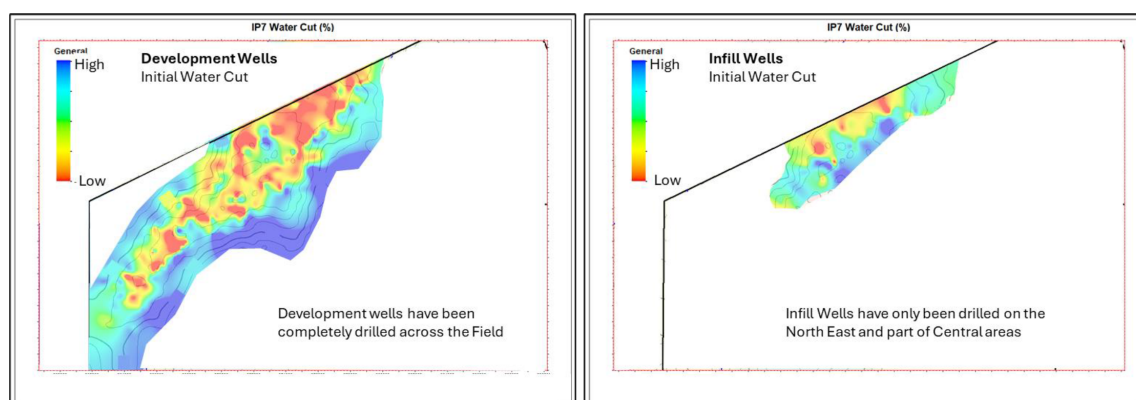
**Figure 2—Pressure mapping and depletion diagnostics indicate faster pressure decline in the Central-West (CW), Central-East (CE), and North-East (NE) areas, suggesting localized variability in aquifer support and reservoir connectivity across the field.**

Insights from the observed spatial variation in reservoir pressure depletion have prompted the initiation of a program focused on enhancing pressure support in targeted areas, particularly the Central region. This initiative, known as the peripheral injection program, is designed to mitigate localized depletion and maintain reservoir pressure with minimal risk of accelerating water production. Beyond stabilizing pressure,

the program aims to achieve a more balanced pressure distribution across the reservoir and ultimately improve overall recovery.

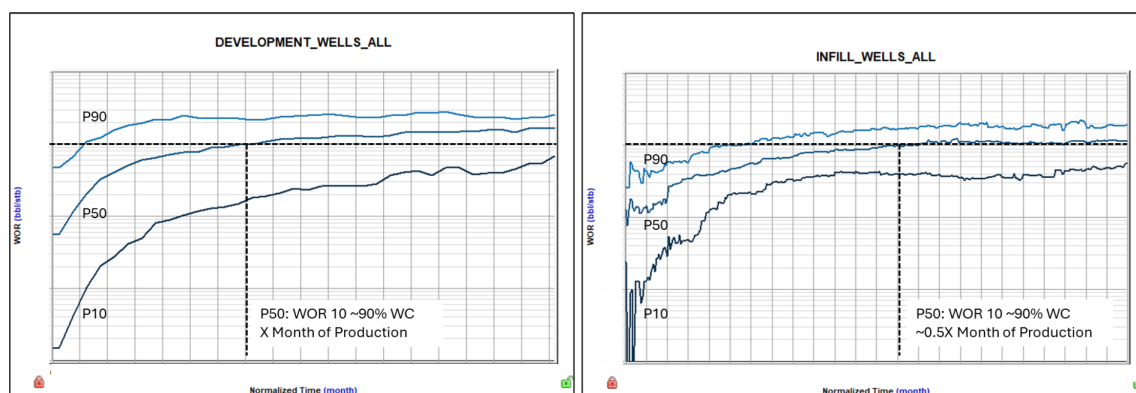
### Water Cut Behavior Mapping

Initial water cut maps were generated by calculating the ratio of cumulative water to total liquid production over the first seven days of production. This metric captures near-wellbore mobile water presence and potential high-permeability streaks that influence early water breakthrough—though distinguishing between the two can be difficult. The maps were further grouped by development wells, infill wells, and a combined dataset to evaluate water cut behavior across different development phases and track the response to infill activity. The maps highlight spatial variation in initial water cut across the field, both in development and infill wells. Development wells reveal certain areas with consistently low initial water cut (mainly in North-East area), while others show early water presence. Infill wells mapping is currently limited, which most of the wells are drilled in the North-East and some in Central areas. The infill wells initial water cut mapping also shows some variation, which some of the North-East area still has low initial water cut.



**Figure 3—Initial water cut maps for development wells (left) and infill wells (right) show spatial variations in early water production. Development wells have been drilled across the entire field, while infill wells are currently limited to the North-East and parts of the Central region.**

The progression of water cut over time indicates that most wells, both development and infill, reach steady-state water cuts of approximately 90%. Infill wells reach this level ~2 times quicker compared to its offset development well. While cumulative-based analysis is preferred for volume-based insights, the uniformity in gross-up procedures across the wells ensures that time-based trends still provide meaningful interpretation. These findings provide an insight into the speed of water cut increment on both types of wells.



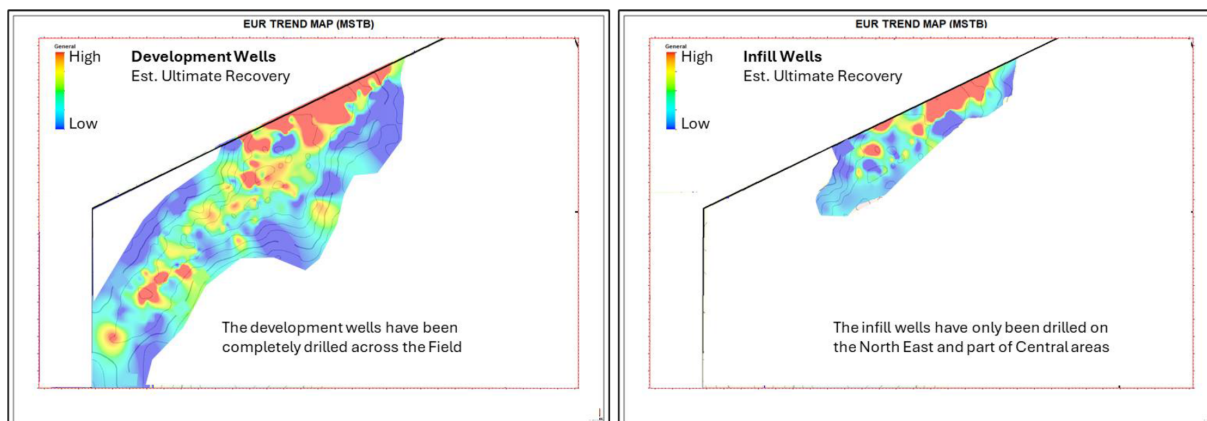
**Figure 4—WOR trends show that development & infill wells reach steady water cut in the level of ~90%. Infill wells reach this steady water cut level ~2 times faster compared to its offset development wells.**



## Estimated Ultimate Recovery (EUR) Mapping

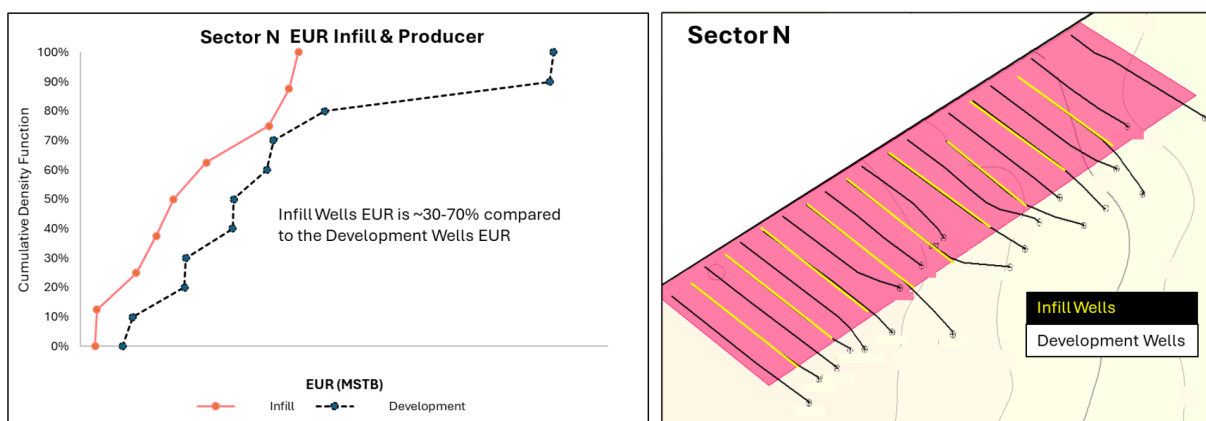
EUR maps were generated using the WOR vs. cumulative production method with a 99% water cut threshold. As with the water cut maps, EUR distributions were analyzed for development wells, infill wells, and the combined dataset. Development wells with lower initial water cut typically correspond to higher EUR, reflecting consistent trends across the field. Areas with favorable EUR remain attractive for further development, while other regions display more heterogeneous results.

Infill well analysis reveals that certain locations maintain relatively high EUR values despite showing elevated initial water cut early in their production, indicating slower water cut progression, extended production life, and greater ultimate recovery. In general, infill well EUR tends to correlate with the performance of their offset wells, where higher offset EUR often leads to more favorable infill outcomes. EUR maps for both development and infill wells show that, although many infill wells have already been drilled—particularly in the North-East and Central regions—the resulting recoveries remain relatively high. This suggests that these areas still hold untapped potential and have not yet reached their economic drainage limits. The sustained strong performance of existing infill wells raises a key consideration: whether additional tightening of well spacing in these regions could be technically and economically justified to further enhance recovery, especially if surface facilities can accommodate the increased production.



**Figure 5—Estimated Ultimate Recovery (EUR) maps for development and infill wells highlight spatial differences in well performance. While development wells have been drilled across the field, infill wells are currently concentrated in the North-East and parts of the Central region—showing favorable EUR trends in targeted areas.**

Evaluation in one of mini sector (Sector N) shows that infill wells tend to recover approximately 30 - 70% of the EUR of their offset development wells. Although the analysis is limited to one area, it provides an early indication of the relative magnitude of infill well performance compared to their offsets. Further assessment confirms that infill wells contribute incremental recovery without degrading the EUR of nearby development wells. The observed well placement, with infill wells drilled between the development wells, demonstrates effective use of spacing reduction to unlock additional volumes. These insights provide a reference for future planning.

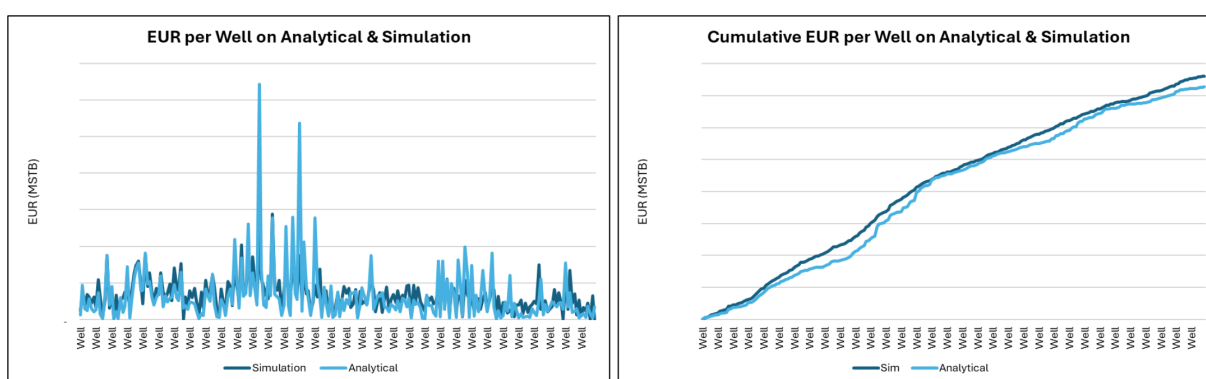


**Figure 6—EUR benchmarking in Sector N indicates infill wells recover ~30 to 70% of their offset development wells (left figure), Sector N that consist of development and infill wells (marked with black & yellow, respectively).**

### Comparison with Simulation

The full-field simulation model was calibrated using production and pressure data up to year-end 2023. The availability of more recent data offers a valuable opportunity to assess model predictability and inform the next modeling cycle. The key focus of the comparison is on well performance metric such as EUR and average reservoir pressure.

In terms of EUR, the reservoir simulation model demonstrates strong consistency with analytical projections for development wells, indicating robust calibration against long-term production behavior. The close match observed at both the individual well level and in aggregate trends suggests the model accurately reflects reservoir dynamics under the development wells phase. The deviation in cumulative EUR between analytical diagnostics and simulation output is approximately ~4%, falling within acceptable bounds of uncertainty for comparative methodologies. However, extension of this evaluation to infill wells is subject to greater uncertainty due to the relatively short production history and limited data density.



**Figure 7—Development Wells EUR per well from both analytical and simulation approaches show strong alignment at the individual well level. The cumulative production results from both methods also exhibit close agreement in total recovered volumes, with discrepancies remaining within an acceptable range.**

Comparison of actual and simulated pressure maps for February 2025 highlights notable pressure discrepancies across the field. In the North-East and South-West crest areas, actual pressures are approximately 100 - 300 psi higher than simulation, indicating that the reservoir holds more energy than predicted. This suggests that aquifer support or transmissibility may be underrepresented in the current model. In contrast, the South Central flank shows around 200 psi lower actual pressure, possibly reflecting overestimated support in that region. Overall, the presence of higher-than-expected pressure in key areas presents an opportunity to leverage the available reservoir energy to further enhance recovery—potentially

through strategies such as increasing the field's gross processing capacity. Unlocking this potential through capacity expansion or optimized offtake could help extend plateau production and improve long-term recovery.

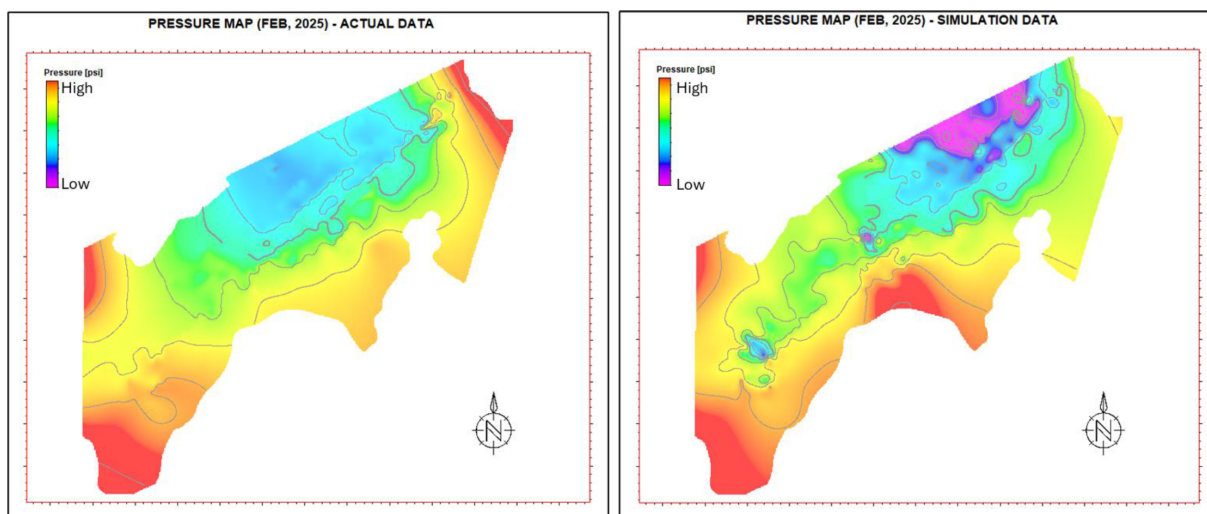


Figure 8—The pressure discrepancy map highlights regions with higher actual pressure than predicted—particularly in the North-East and South-West crest.

### Action Derived from Results: Infill Wells Optimization

Based on the observed discrepancies in water cut, favorable EUR trends, and higher-than-expected reservoir pressure, an infill well optimization was carried out. The effort focuses on revising the drilling sequence—prioritizing high-potential locations while deferring lower-performing candidates. Given the constraints in water handling capacity, advancing better-quality wells helps average down field-wide water cut and maximize overall oil production. This approach also ensures that limited capacity is allocated to the most efficient wells, improving early-cycle value. The updated sequencing is continuously refined using well performance diagnostics and offset analysis to guide decision-making.

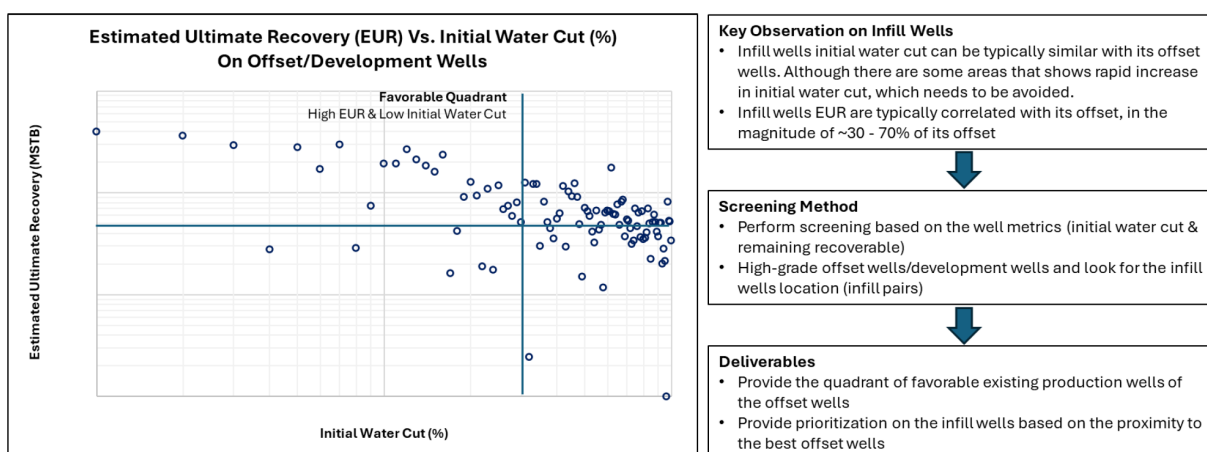


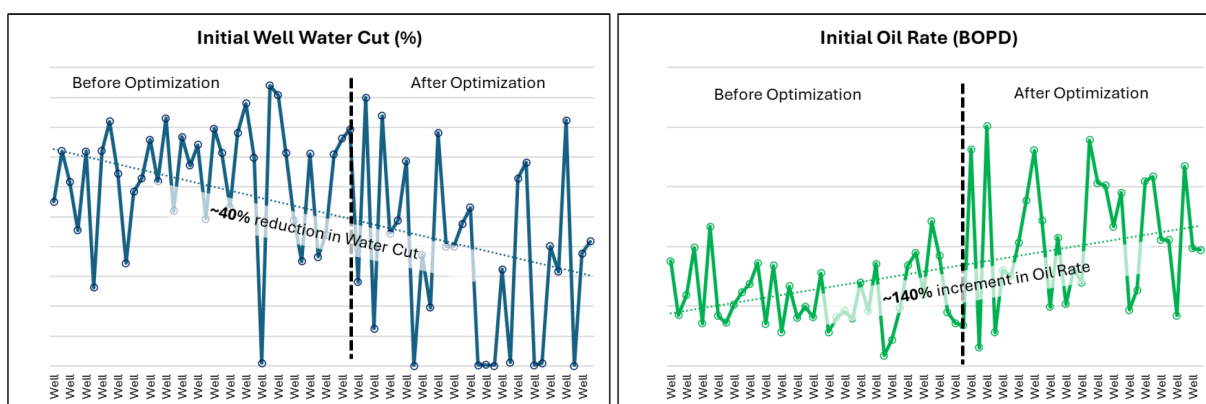
Figure 9—A log-log plot of offset well EUR versus initial water cut highlights a favorable quadrant in the top-left region, representing wells with high EUR and low initial water cut. This insight supports prioritizing infill wells candidates based on the performance of their corresponding offset wells.

The main changes in process of the existing drilling sequence are reallocation and accelerated drilling in areas showing favorable facies to optimize productivity. Several initiatives that have been carried out utilizing these insights such as:



- Proposed new infill wells in the North-East area.
- Accelerated drilling in South-West area due to identification of localized good facies.
- Re-allocated wells with directional changes to enhance productivity.
- Delay some wells that have poor offset performance.

This targeted approach resulted in an average 40% reduction in water cut, reflecting the effectiveness of selecting and sequencing wells based on favorable reservoir conditions. Concurrently, new well initial oil production rates increased by approximately 140%, demonstrating that prioritizing high-quality zones led to early gains in hydrocarbon output. The observed improvements are a direct outcome of optimizing the drilling sequence to capture the most productive intervals earlier in the development timeline. The results underscore the value of integrating analytical diagnostics into infill planning workflows to guide operational execution. By aligning subsurface insight with development timing, the program effectively enhanced both production performance and system efficiency.



**Figure 10—Infill well prioritization based on initial water cut and EUR analysis led to ~40% reduction in water cut and ~140% increase in oil rate, demonstrating the effectiveness of drilling high-quality targets earlier in the development sequence.**

### Action Derived From Result: Refine Geological Insight

Analytical results are used to provide feedback to geoscientists for the refinement of the geological model. One key application is in improving property distribution, where the EUR trend map is utilized as a secondary attribute to guide porosity distribution across the field. This approach integrates dynamic performance data into the static modeling workflow, allowing the porosity model to better reflect zones of observed productivity rather than relying solely on well log or core data. The alignment of high EUR regions with elevated porosity values enhances geologic realism and predictive capability of the model. The porosity model is among several parameters to be tested against production attributes. The objective is to identify which dynamic parameters best align with geological insights and can support further refinement of the model.

The feedback loop from dynamic modelling and production data analysis into the static model is always a key aspect of any modelling workflow, and even more critical in complex heterogeneous carbonate reservoirs. With the established analytical and simulation-based analysis workflow a process of continued improvement can be adopted with repeated feedback to the static model supported by increased availability of new wells and production data. Strengthening of existing trends and development of new ones can be used to further enhance geological understanding and ultimately improve the ability of models to forecast future production, enhancing development decisions.

### Action Derived From Result: Maximize Field Recovery

The analysis highlights opportunities to increase recovery through well sequence optimization, tighter spacing, and additional capacity. Sequence prioritization based on offset performance is becoming increasingly important as the inventory of high-quality drilling locations diminishes. The sustained performance of certain infill wells with low initial water cut and high recovery supports further evaluation of tighter well spacing as a viable strategy. However, this approach is only effective if paired with sufficient facility capacity to handle high water production—currently steady at 90–95% field water cut. These opportunities will be further assessed through technical and economic evaluations to determine the feasibility of an additional development phase.

## Conclusions

This study demonstrates that combining classical reservoir engineering diagnostics with dynamic simulation and geoscience integration enhances the understanding of reservoir behavior and supports informed decisions for infill drilling optimization in mature carbonate reservoirs. The key conclusions drawn from this integrated analysis are:

- Water influx was confirmed as the primary reservoir drive, maintaining pressures at 70-90% of initial values. Spatial variation in pressure depletion suggests more restricted vertical communication between the aquifer and productive zones in certain areas.
- Initial water cut exhibits spatial variability across the field in both development and infill wells. Water cut progression analysis indicates a stable trend toward 90%, infill wells reach the stable water cut ~2 times faster compared with development.
- EUR mapping reveals spatial variability in recovery performance across both development and infill wells. Analysis within a representative sector indicates that infill wells typically achieve ~30 - 70% of the EUR of their offset development wells.
- Comparison between analytical results and simulation highlights good agreement in EUR for development wells, confirming the model's reliability in capturing long-term recovery trends.
- Implementation of a revised infill drilling strategy, focused on prioritizing areas that exhibit lower initial water cut, successfully reduced early water production by approximately 40% while also increasing the initial oil rate of infill rate by approximately 140%.
- Insights from the evaluations are used to refine the geological understanding of the area, with the production trend map now applied as a secondary parameter to guide more representative static parameter distribution.
- Opportunities to maximize recovery through well sequence optimization, tighter spacing, and additional capacity remain viable, particularly in low water cut areas, but require careful technical and economic evaluation given the high field-wide water cut.

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